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Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

Session	**2025-26 (Even)**	**Course Name**	**PE-III: Lab -Reinforcement Learning**
Semester	**6th**	**Course Code**	**23ADS1644**
Roll No	**42**	**Name of Student**	**Krishna Mishra**

Practical Number	**02**
Course Outcome	CO1: Apply fundamental reinforcement learning concepts to model and solve decision-making problems. CO2: Implement basic reinforcement learning algorithms to train agents in a given environment. CO3: Analyze agent behavior
Aim	To explore various Python libraries for local data import, data cleaning, preprocessing, and visualization techniques.
Problem Definition	The objective of this experiment is to explore Python libraries such as Pandas, Matplotlib, and Seaborn for importing local datasets, performing basic data cleaning and preprocessing, and visualizing data using different graphical techniques to gain meaningful insights.
Theory (100 words)	Data visualization and preprocessing are essential steps in data analysis that help in understanding patterns, trends, and relationships within a dataset. Python provides powerful libraries such as Pandas for data manipulation, Matplotlib for basic plotting, and Seaborn for advanced statistical visualizations. Data preprocessing involves handling missing values, selecting relevant features, and preparing data for analysis. Visualization techniques like histograms, box plots, scatter plots, bar charts, and heatmaps allow users to interpret distributions, detect outliers, and identify correlations between variables. The Iris dataset is widely used for learning these concepts, as it contains structured numerical data across different species. Using these libraries in environments like Google Colab enhances efficiency and enables interactive data exploration and analysis.

Procedure and Execution (100 Words)	Algorithm: 1. Import required libraries: Pandas, Matplotlib, and Seaborn. 2. Load the dataset (Iris.csv) using pd.read_csv(). 3. Set Seaborn style for better visualization. 4. Plot a histogram of Sepal Length to observe distribution. 5. Create a box plot of Petal Length to detect outliers. 6. Generate a scatter plot between Sepal Length and Petal Length with species differentiation. 7. Plot a count plot to show frequency of each species. 8. Create a heatmap to visualize correlation between numerical features.

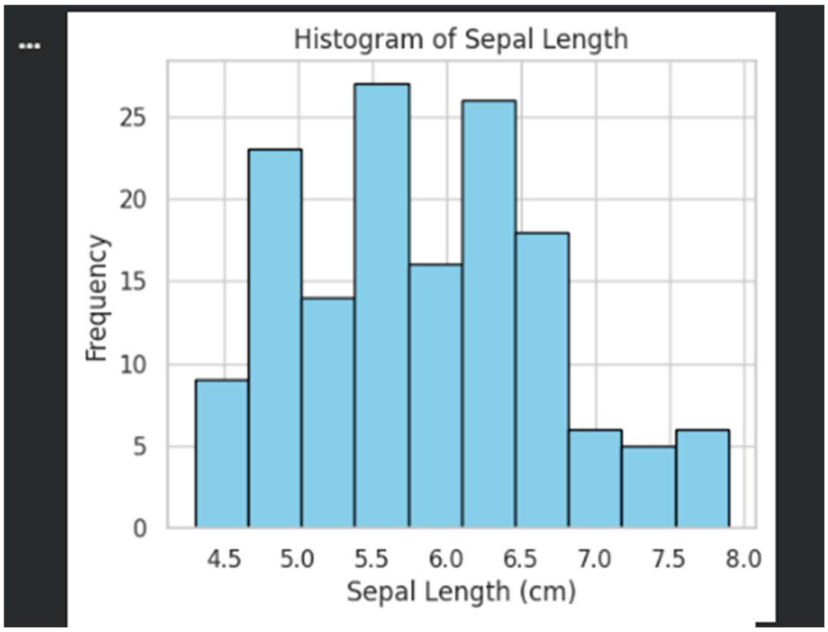
Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv("Iris.csv")
sns.set(style="whitegrid")

plt.figure(figsize=(5,4))
plt.hist(df['SepalLengthCm'], bins=10, color='skyblue', edgecolor='black')
plt.title("Histogram of Sepal Length")
plt.xlabel("Sepal Length (cm)")
plt.ylabel("Frequency")
plt.show()

plt.figure(figsize=(5,4))
sns.boxplot(y=df['PetalLengthCm'])
plt.title("Box Plot of Petal Length")
plt.ylabel("Petal Length (cm)")
plt.show()

plt.figure(figsize=(5,4))
sns.scatterplot( x='SepalLengthCm',  y='PetalLengthCm', hue='Species',
data=df )
plt.title("Scatter Plot: Sepal Length vs Petal Length")
plt.xlabel("Sepal Length (cm)")
plt.ylabel("Petal Length (cm)")
plt.show()
```

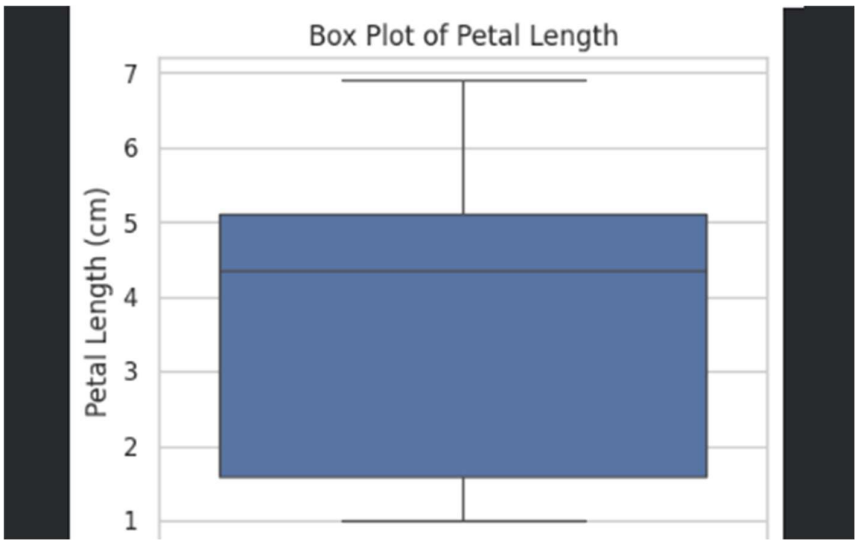
```
plt.figure(figsize=(5,4))
sns.countplot(x='Species', data=df)
plt.title("Bar Chart of Iris Species Count")
plt.xlabel("Species")
plt.ylabel("Count")
plt.show()

plt.figure(figsize=(6,5))
sns.heatmap(
    df.drop(columns=['Id', 'Species']).corr(),
    annot=True,
    cmap='coolwarm' )
plt.title("Correlation Heatmap")
plt.show()
```

Output:

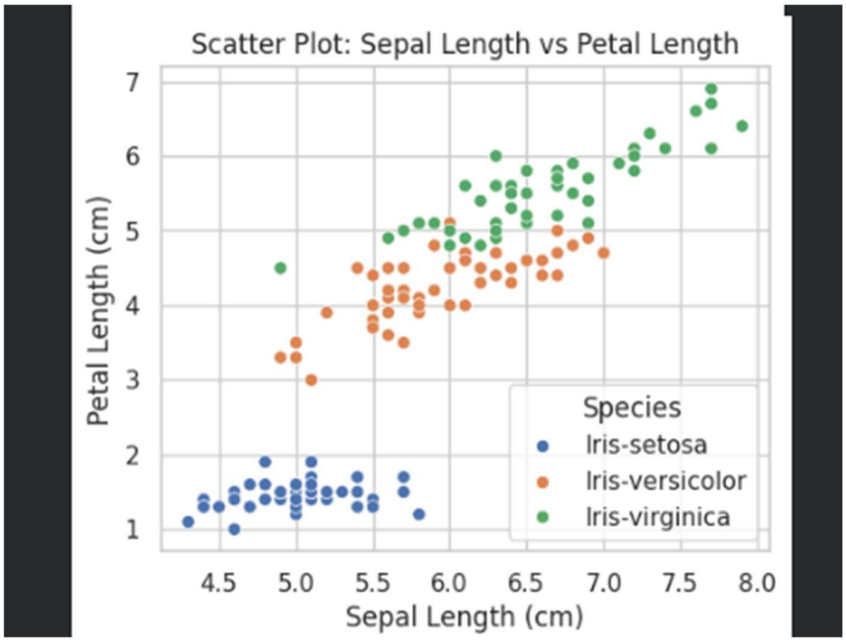
1. **Histogram**



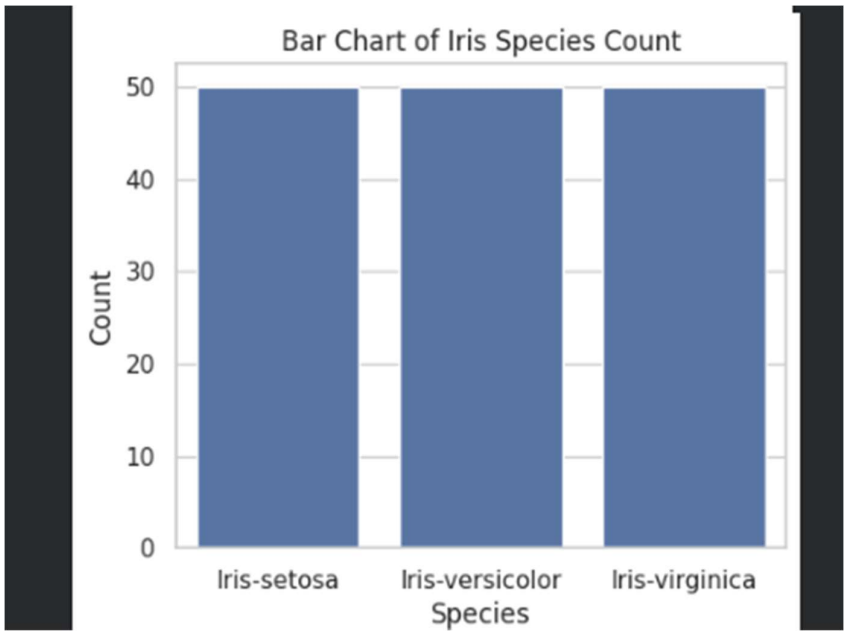
2. Box Plot



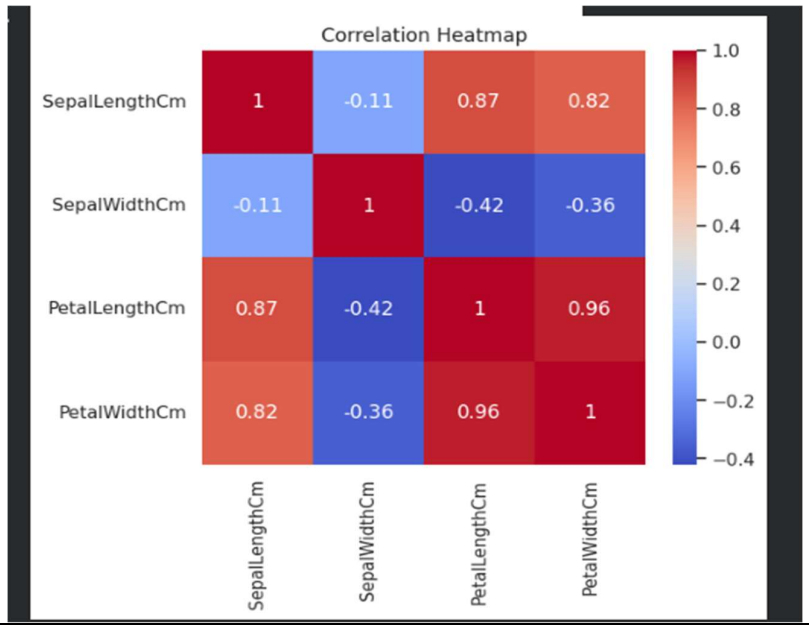
3. Scatter Plot

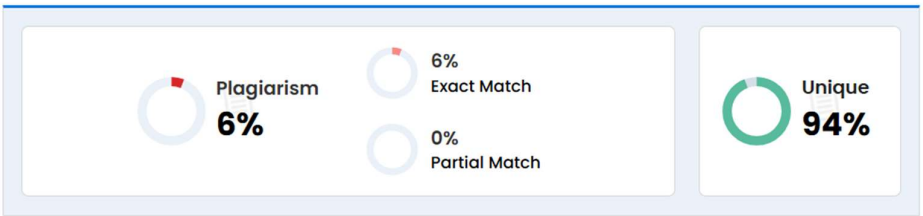


4. Bar Plot



5. **Heatmap**



Output Analysis	• The histogram shows the distribution of sepal length, indicating how values are spread. • The box plot highlights median, quartiles, and possible outliers in petal length. • The scatter plot reveals relationships between sepal and petal length and shows clear clustering among species. • The count plot displays the number of samples in each species category, confirming dataset balance. • The heatmap shows correlation values between features, helping identify strong relationships (e.g., petal length and petal width are highly correlated).
Link of student Github profile where lab assignment has been uploaded	https://github.com/krish-na45/RL_Practicals
Conclusion	This experiment demonstrates how Python libraries can be effectively used for data import, preprocessing, and visualization. The visualizations help in understanding data distribution, detecting outliers, and identifying relationships between variables. Such analysis is essential for making informed decisions and preparing data for machine learning models.
Plag Report (Similarity index < 12%)	
Date	28/03/2026